

KANIKA RAWAT

PhD

Indian Institute of Science, Bengaluru

Email: kanikarawat@iisc.ac.in; kanikarawat59@gmail.com

EDUCATION

- **PhD Biology**, Centre for Ecological Sciences, IISc Bengaluru (2026); CGPA: 8.1
Coursework: Evolutionary Biology; Animal Behaviour; Plant-animal Interactions; Mathematics and Statistics for Biologists; Quantitative Ecology; Process of Scientific Thinking
 - **MSc Biotechnology**, Department of Microbiology and Biotechnology Centre, Maharaja Sayajirao University Baroda (2018); CGPA: 8
 - **BSc Zoology**, Miranda House, Delhi University (2016); CGPA: 8.5
 - **Senior Secondary (Class XII)**, K.V. ONGC Dehradun, CBSE (2013); CGPA: 9
-

RESEARCH EXPERIENCE

Research Associate | NCBS Bengaluru | 2025-present

I am **collaborating** with ecologists, molecular biologists, and theoreticians to model zoonotic spillovers in Northeast India, informing **sampling and control strategies**.

PhD | IISc Bengaluru | 2019-2026

I **co-led** a research group of five people, along with my supervisor, **Dr Kavita Isvaran**. We studied the **ecology** and life-history **evolution** of the ***Aedes aegypti*** mosquito, with a specific focus on the understudied pupal stage. To link early ecological conditions with the **disease transmission potential** of a female mosquito, we performed **controlled behavioural, life-history and biochemical experiments**.

MSc | MSU Baroda | 2017-2018

I worked with **Dr Rajesh Patkar** (IIT Bombay) and **Dr Hiral Shah** (EMBL) at M.S. University Baroda. We studied the dynamics of **kinetochore proteins** in the pathogenic **rice blast fungus**. Our aim was to find the potential targets to stop its growth in the plant. We **fluorescence-tagged** the protein genes and studied their localization using **confocal microscopy**.

Conferences and Workshops

- Presented my PhD work at **Behaviour**, Kolkata (2025); **ISBE** (International Society for Behavioral Ecology), Melbourne (2024); Stockholm (2022); and **ESEB** (European Society for Evolutionary Biology), Prague (2022)
- Selected for ICTS program on **Emerging infectious diseases: ecology and evolution** (July 1, 2024 to July 12, 2024) at ICTS-TIFR, Bengaluru

Publications

- **Rawat, K.**, & Isvaran, K. (2026). Peers over the past: prior predation-risk experience does not dictate antipredator responses of individuals in groups. *Biology Letters*, 22(4). <https://doi.org/10.1098/rsbl.2025.0806>
 - **Rawat, K.**, Bhambore, A. A., & Isvaran, K. (2025). Don't leave the past behind: larval experience shapes pupal antipredator response in *Aedes aegypti*. *Behavioral Ecology*, 36(2). <https://doi.org/10.1093/beheco/araf001>
 - Shah, H., **Rawat, K.**, Ashar, H., Patkar, R., & Manjrekar, J. (2019). Dual role for fungal-specific outer kinetochore proteins during cell cycle and development in *Magnaporthe oryzae*. *Journal of Cell Science*, 132(12). <https://doi.org/10.1242/jcs.224147>
 - **Rawat, K.**, Francis, N. M., Samaddar, S., & Isvaran, K. Insights into the life-history evolution of *Aedes aegypti* via the lens of multistage trade-off. (*in prep*).
-

TECHNICAL SKILLS

- **Disease modelling**– compartment and agent-based models
 - **Technical writing and editing**
 - **Study and experiment design**– identifying questions, generating hypotheses, testing predictions using standardized experimental or observational data
 - **Data management, analyses, visualization, and interpretation**
 - **Statistics in R**– general and generalized linear models (Negative binomial, Poisson, Beta, and Multinomial regression), mixed-effects models, bootstrapping in R and ggplot
 - **Molecular techniques**– DNA extraction, PCR, Gel electrophoresis, DNA cloning, Southern and Western hybridization
-

ACADEMIC ACHIEVEMENTS

- **Cover image** selected for the May 2026 issue of *Biology Letters*
- Received **grants** to present PhD work at ISBE Congress Turin (2026), Gordon Research Conference on Predator-Prey Interactions (2026), Behaviour Congress Kolkata (2025), ISBE Congress Melbourne (2024) and Stockholm (2022), and ESEB Congress Prague (2022)
- **Co-wrote a grant proposal** with Dr Kavita Isvaran, successfully securing a SERB-SUPRA grant (2019)
- Selected for **PhD Program** in Biology at **IISc** Bengaluru, **IISER** Pune, and **JNCASR** Bengaluru (2018)
- Secured ranks in **national-level exams** (2018):
 - Council of Scientific and Industrial Research– JRF, rank 35
 - Department of Biotechnology– JRF Programme, Category I- rank 34
 - Graduate Aptitude Test– Life Sciences, All India Rank- 44

- **Gold-medal**, MSc Biotechnology, MSU Baroda (2018)
-

CO-CURRICULAR CONTRIBUTIONS

- **Taught** animal behaviour to bachelor's, master's, and PhD students as a **Guest Lecturer at IISc** (2025)
 - **Trained** eight bachelor's students and five project assistants in study design, behavioural observations, data entry, mosquito handling, and experiments (2020-2024)
 - **Mentored** three master's and two bachelor's theses (2022-2023)
 - **Taught** BSc and PhD students and **assisted** with a course on study design, statistics, and R programming (2021)
 - **Organized** invited talks and **volunteered** for workshops (2019-2023):
 - Inclusive Ecology online workshop
 - In-house symposium at Centre for Ecological Sciences, IISc
 - Student Conference on Conservation Science, Bengaluru
-

EXTRA-CURRICULAR ACHIEVEMENTS

- **Team captain** of IISc **Ultimate frisbee** team (mixed-gender team) (2023)
 - **Silver medal** at National College Ultimate Championship (2022)
 - Qualified **Indian Air Force** SSB and Pilot Aptitude Battery Test (2016)
 - **Gold medal** at National Level Western Group **Dance** Competitions at BITS Pilani, Indian School of Business and Finance, and St. Stephen's College (2014-2015)
-